Vinayak Ichake

M.Tech - Computer Science and Engineering
Indian Institute Of Technology, Guwahati

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Vinayak-Ichake

Education

Degree	Institute/Board	CGPA/Percentage	Year
M.Tech.	Indian Institute of Technology, Guwahati, Assam	8.71	2023-2025
B.E.	Government College of Engineering and Research, Awasari Khurd	8.34	2018-2022

Experience

- **Axtria Ingenious Insights** *May 2024 - July 2024*
  Analyst Intern Pune, Maharashtra
  – Leveraged **SQL** for data validation and managing data flows within MDM and ETL processes, while utilizing **Jira**, **Snowflake**, and Axtria products for data integration and advanced analytics.
- **HCL Technologies Ltd.** *Sept 2022 - July 2023*
  Graduate Engineer Trainee Pune, Maharashtra
  – Engineered full-stack solutions in a collaborative team, using **Java/Spring Boot** for backend services and **ReactJS** for frontend components, resulting in a **5%** boost in application performance .

Projects

- **Scalable IBS-based Blockchain Framework for IoT Transaction Management | M.Tech. Thesis** *June 2024 - May 2025*
  Prof.Sukumar Nandi, Professor, CSE ,IIT Guwahati
  – Pioneered a scalable **Identity-Based Signature (IBS)** blockchain framework for IoT Transaction Management, leveraging the Barreto-Libert-McCullagh-Quisquater (BLMQ) scheme to achieve low latency up to **52ms**.
  – Engineered a decentralized, region-based architecture that significantly enhanced IoT scalability and throughput, reducing memory consumption by over **80%** compared to traditional setups
  – Tools and Technologies: Hyperledger Fabric, Blockchain, BLMQ Algorithm, IPFS, ERC-721
- **Detection of Dyslexia Using Handwriting Images** | **Course Project** | *Mar. 2024 - May 2024*
  Debanga Raj Neog, Assistant Professor, MFSDS&AI, IIT Guwahati
  – Designed an automated system to detect dyslexia by analyzing handwriting, using distinct writing patterns.
  – Compared performance across classifiers (**ANN**, **SVC**,**Decision Tree**) with features derived from both images and text, demonstrating superior results with image-specific features.
  – Tools: OpenCV2, Tesseract,TensorFlow.
- **Multi-threaded Event Reservation System** | **Course Project** | *Mar. 2024 - May 2024*
  Dr. Arijit Sur, Professor, Dept. of CSE, IIT Guwahati
  – Implemented a concurrent reservation system using **pthread API**,employing mutual exclusion and thread synchronization to ensure data consistency.
  – Optimized event booking, cancellation, and inquiry operations by **25%**.
  – Tools: C
- **Speech-based Digit Recognizer** | **Course Project** | *Sept. 2023 - Nov. 2023*
  Dr. Pradip K. Das, Professor, Dept. of CSE, IIT Guwahati
  – Implemented a speech-based digit recognition system employing the **Hidden Markov Model(HMM)** to enable the identification and classification of spoken digits with a demonstrated accuracy of **95%.**
  – Tools: C++, Visual Studio, Cool Edit.

Technical Skills

- **Programming Languages**: C/C++, Python , Java , Javascript
- **Frameworks and Libraries**: Spring Boot, ReactJS *
- **Miscellaneous**: MySQL, VS code, Git, Docker, RESTful API
- **Operating Systems**: Windows, Linux

* *Elementary proficiency*

Key courses taken

- Speech Processing , Image Processing with Machine Learning, Artificial Intelligence
- Algorithms and Data Structures , Operating Systems , Database Management System

Extra-Curricular Activities

- **Workshop on Blockchain Tools and Technologies**,
  Conducted a comprehensive, hands-on workshop on Hyperledger Fabric, providing 500+ students with a foundational understanding of private blockchain implementation.

Achievements

- **GATE CS**, All India Rank **983** among 75680 candidates in GATE 2023
- **Algorithmic Problem-Solving**,Solved over 1000+ challenges on LeetCode (Contest Rating: **1950(Knight)**.